



Ancient subsurface life

The microbes living deep below the ground may be independently evolving survivors of ancient life forms. <http://digit.in/sep20-49>



Games helping the planet

Videogames are mobilising their legions of fans in various efforts to lend a helping hand to Mother Earth. <http://digit.in/sep20-50>

in conditions of extreme pressure and temperature. Since then, there have been many researches into the deep biosphere, mostly using mines, oil fields, or samples drilled using research vessels. One species, a single cell organism called *Geogemma barossii*, can thrive in hydrothermal vents at temperatures higher than the boiling point of water, at 121 degrees Celsius. Another species, *Candidatus Desulforudis* was found in isolation from all other microbial life, being the first known organism to form an ecosystem all on its own! The organisms have been found beneath the continental crust, as well as the oceanic crust. They occupy the biggest habitat on the planet. In recent years, we have understood much about the deep biosphere because of the studies by the Deep Carbon Observatory and the Deep Life community, an international collaboration of many universities and scientists across disciplines, with the overall aim of understanding the role of carbon in the life processes on the planet.

These studies, which took place over a decade, provided some surprising insights about the deep biosphere. The biggest revelation was the sheer amount of these creatures. It is estimated that these organisms account for between 15 to 23 billion tons of carbon, hundreds of times more than all the humans on Earth put together. Bacteria and archaea living in just the crust beneath the oceans, weigh as much as all the life that lives within the oceans. Researchers now estimate that almost 70 percent of microbial life are subterranean, sometimes also

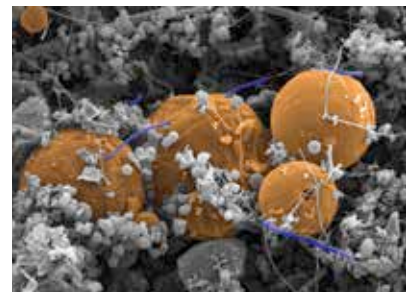
known as intraterrestrial life, and most of these species are little studied and not characterised. As much about them is still unknown, they make up what scientists are referring to as “microbial dark matter”. The few studies conducted on them indicate that their genetic diversity matches that of life on the surface, and may even exceed it.

No matter how deep researchers have probed, they have found life there, even at depths in excess of 2,000 meters beneath the seafloor. These bacteria were producing methane from coal deposits that were tens of millions of years old. Unlike the bacteria living in relatively shallower depths of the crust, the bacteria in the coal deposits were similar to their cousins living in jungle ecosystems. This makes sense considering that the coal originates from buried forest-like environments. The buried bacteria are actually relatives of creatures that lived in shallow coastal jungles millions of years ago. The record depths at which organisms are found are 5 km below land, and 10.5 km below oceans.

Deep within the ground, the life processes are slow, and the organisms are ancient. It may well be possible that the oldest living organisms is not some tree, but some microbe deep within the ground. Deep within the ground, the rate of metabolism of these organisms is incredibly slow, and they breathe with almost infinite patience, much slower than their cousins closer to the surface at least. In a lab, the more regular forms of microbial life have extremely fast life cycles and can reproduce once every 20 minutes or faster. Deep within the ground though, the life processes are slowed down and cell division may not take place for 100 years or longer. These are not the microbes in suspended animation but the living, breathing microbes which still are hydrated and have access to water.

In the Deccan Traps, which are the eroded remains of massive volcanic eruptions 65 million years ago, researchers from IIT Kharagpur and the University of California San Diego,

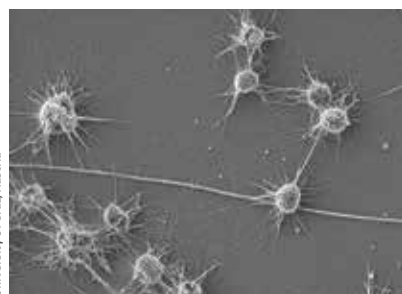
USA collected samples of rock to investigate the bacterial in the volcanic rock, as well as the granite bedrock which is even older, as much as 2.5 billion years old. These are volcanic rock with low nutrients and the pressure increases by more than 250 times than the pressure at sea level, for every 1000 meters of rock. In the basalt, the temperature



The purple rod shaped cells between the orange blobs of carbon are *Amdaxiobacter*, a species that is found in isolation 2.8 km beneath the surface

also rises by 25 degrees Celsius for every 1000 meters. The archaea living here were extremely hardy, resistant to acid, heat and metabolising iron and sulfur. These populations have existed in the region from before the volcanic activity 65 million years ago. Some of the organisms are related to marine microbes, which indicate that at some ancient point in time, seawater may have infiltrated deep underground. At the Kidd Creek Mine in Canada, researchers have identified bacteria living in pockets of water that were isolated 2.7 billion years ago – the oldest known reservoir of groundwater on the planet.

There are still many questions that remain to be answered. For example, we do not know if life originated on the surface, or crawled upwards towards the sunlight from deep within the Earth. We do not know how life forms within the Earth spread or move from place to place. How geological activity such as plate tectonics, earthquakes and volcanoes affect these life forms is also not well understood. Then there is the matter of the zombie life forms, those that can live for tens of millions of years without reproducing. 



Archaea known as *Altiarchaeales* found in sulfidic springs of Germany

Credit: Christine Moissl-Eichinger, Medical University of Graz, Austria

Credit: Greg Wanger and Gordon Southam